7	(a)		each line represents photon of specific energy	M1	
			photon emitted as a result of energy change of electron	M1	
			specific energy changes so discrete levels	A1	[3]
	(b)	(i)	arrow from –0.85 eV level to –1.5 eV level	B1	[1]
		(ii)	$\Delta E = hc/\lambda$	C1	
			$= (1.5 - 0.85) \times 1.6 \times 10^{-19}$	C1	
			$= 1.04 \times 10^{-19}$ J		
			$\lambda = (6.63 \times 10^{-34} \times 3.0 \times 10^{8})/(1.04 \times 10^{-19})$		
			$= 1.9 \times 10^{-6}$ m	A1	[3]
	(c)		spectrum appears as continuous spectrum crossed by dark lines	B1	
			two dark lines	B1	
			electrons in gas absorb photons with energies equal to the excitation energies	M1	
			light photons re-emitted in all directions	A1	[4]
8	(a)	(i)	time for initial number of nuclei/activity	M1	